

Project 1 Report

Decentralized Robot Fleet Governance

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1. Introduction and Objectives

This project aims to address the vulnerabilities of traditional centralized fleet management systems by establishing a "Decentralized Governance" architecture powered by Blockchain technology and Smart Contracts. A smart contract was developed using Solidity to securely manage robot authentication, task assignment, and outcome verification within a Zero-Trust Environment.

2. Architecture & Programming Logic

To optimize computational load and minimize Gas costs on the blockchain, the following strategies were implemented:

- **Capability Bitmask:** A uint128 variable was utilized to store complex hardware capabilities (e.g., Lidar sensors, heavy payload capacity) as binary digits. This method significantly reduces storage overhead compared to using strings or arrays.
- **Multiplier Gate:** Since Solidity does not natively support floating-point mathematics, fractional sensor readings were scaled by a factor of 10^7 into on-chain integers to maintain precise computational integrity.

3. The Three Security Checkpoints

The system's stability and security rely on routing every command through three exhaustive on-chain logic gates to prevent unauthorized access and operational failures:

Checkpoint 1 (Invalid Robot Validation): Ensures that unvetted or unregistered robots cannot interact with the swarm queue or accept tasks.

Checkpoint 2 (Concurrency Shield): Prevents over-tasking and race conditions by actively rejecting double-booking attempts if a robot is already engaged in an active task ("Robot Busy").

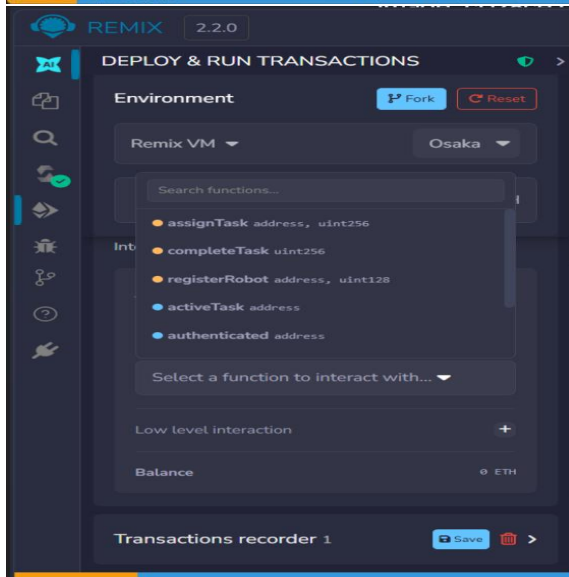
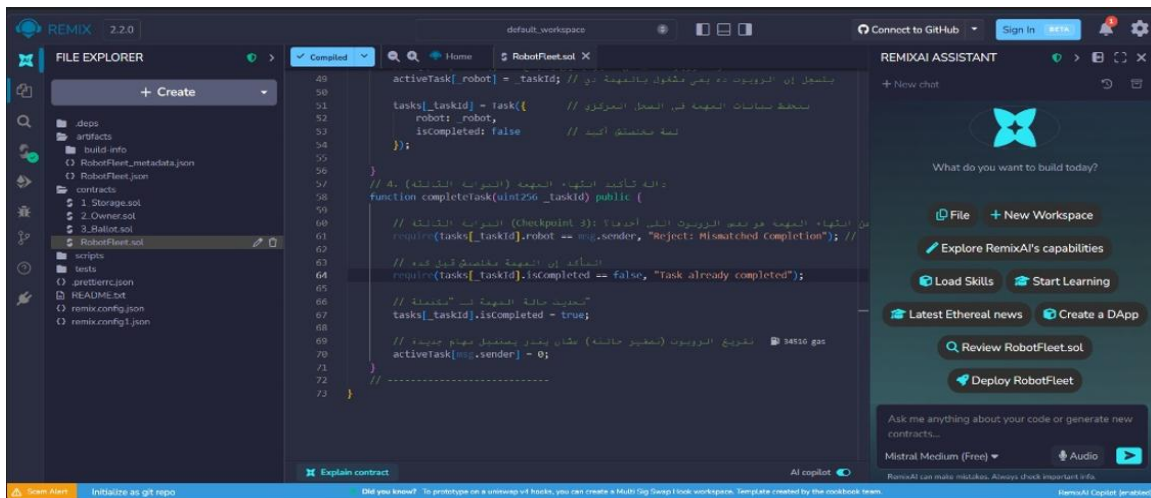
Checkpoint 3 (Mismatched Completion): A strict access control mechanism ensuring that only the specific robot assigned to a task can report its successful completion, thereby preventing unauthorized state alterations.

4. Deployment & Simulation

The smart contract was successfully compiled and deployed on a local Remix VM environment. Comprehensive operational scenarios were simulated—including authenticated registration, exclusive tasking, and proof-driven completion—flawlessly passing all security gates without failure.

5. Attachments: Practical Execution (Screenshots)

Please find the execution proofs attached below:



6. Conclusion

This project successfully demonstrated the transition from fragile hierarchical control to a mathematically precise, economically self-sustaining decentralized network. By shifting decision-making logic on-chain, the developed system establishes a resilient and secure governance layer for autonomous mechanical fleets.